26900079 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOTA05 - BIO-INSTRUMENTATION AND QUALITY ANALYSIS 3

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer in One or Two Sentences

Answer any **10** questions

Each question carries **2** marks

1. What are simple phenolics and polyphenolics? [K] / [CO1]
2. What is pseudoalkaloid? [K] / [CO1]
3. What you mean by secondary structure of RNA? [U] / [CO1]
4. What are polyunsaturated fatty acids? Give an example [U] / [CO1]
5. Explain the extraction chamber of a Soxhlet apparatus [U] / [CO2]
6. What is partition coefficient? How K value of a phytochemical determine solubility [A] / [CO2]
7. Explain Bromine water test for tannin detection [U] / [CO3]
8. How can you quantify DNA using UV spectrophotometer [A] / [CO3]
9. What is DNS reagent? How can this be used in detecting reducing sugar [K] / [CO3]

10. Explain Oil Red O staining of lipids [U] / [CO3]

11. What is the advantage of free drying over rotary evaporation? [U] / [CO4]

12. List out applications of GC MS analysis [A] / [CO4]

[2x10 = 20]

Part B

Short Answer Type Questions

Answer any **8** questions

Each question carries **5** marks

13. What are the functions of flavanoid? [K] / [CO1]

14. Write a note on primary metabolites [K] / [CO1]

15. Explain the biosynthesis of carbohydrate [U] / [CO1]

16. Explain manual grinding and knife-blade-hammer grinding [U] / [CO2]

17. Explain cold maceration method of metabolite extraction [U] / [CO2]

18. Explain Foam test and hemolysis test for saponin detection [U] / [CO3]

19. Explain the principle and procedure of Bradford's method of protein determination [U] / [CO3]

20. Explain Hager's test and Mayer's test for the detection of alkaloids [U] / [CO3]

21. What is solvent-solvent extraction? Explain co-current method. [U] / [CO4]

22. Explain IR spectrophotometer [U] / [CO4]

[5x8 = 40]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

23. Write an essay on Ultrasound assisted extraction and Super Critical Fluid extraction [K] / [CO2]

24. Explain vertical gel electrophoresis [U] / [CO4]

[10x1 = 10]